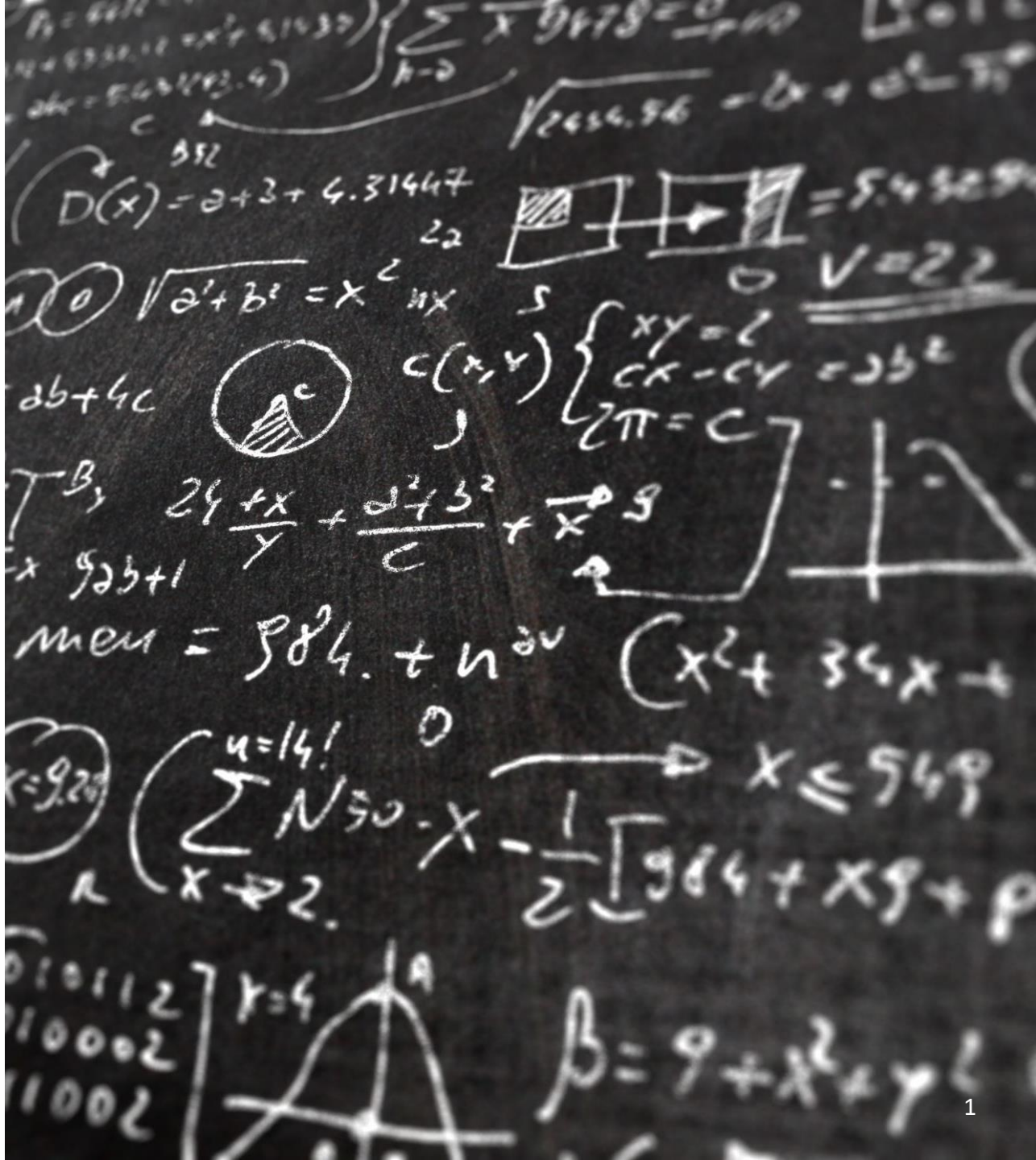
Chapter 2: Equations



1. linear equation

A linear equation is an algebraic equation where the highest power of any variable is always one, and its graph makes a straight line.

Linear Equations in One Variable

A linear equation is an equation of a straight line, written in one variable. The only power of the variable is 1. Linear equations in one variable may take the form $ax + b = 0$ and are solved using basic algebraic operations.

linear equation in one variable

A linear equation in one variable can be written in the form

$$ax + b = 0$$

where a and b are real numbers, $a \neq 0$.

Th **solution set** consists of all values that make the equation true. For this equation, the solution set is all real numbers because any real number substituted for x will make the equation true.

An **identity equation** is true for all values of the variable. Here is an example of an identity equation. $3x = 2x + x$

A **conditional equation** is true for only some values of the variable. For example

$$5x + 2 = 3x - 6$$
$$2x = -8$$
$$x = -4$$

The solution set consists of one number: $\{-4\}$. It is the only solution

Example: Solve the following equation: $2x + 7 = 19$

$$2x + 7 = 19$$
$$2x = 12$$
$$x = 6$$

Example: Solve the following equation: $4(x - 3) + 12 = 15 - 5(x + 6)$

$$4(x - 3) + 12 = 15 - 5(x + 6)$$
$$4x - 12 + 12 = 15 - 5x - 30$$
$$4x = -15 - 5x$$
$$9x = -15$$
$$x = -15/9$$
$$x = -5/3$$

Example : Solve the rational equation: $\frac{7}{2X} - \frac{5}{3X} = \frac{22}{3}$

$$(6x)\left(\frac{7}{2x} - \frac{5}{3x}\right) = \left(\frac{22}{3}\right)(6x)$$

$$(6x)\left(\frac{7}{2x}\right) - (6x)\left(\frac{5}{3x}\right) = \left(\frac{22}{3}\right)(6x)$$

$$(\cancel{6x})\left(\frac{7}{\cancel{2x}}\right) - (\cancel{6x})\left(\frac{5}{\cancel{3x}}\right) = \left(\frac{22}{\cancel{3}}\right)(\cancel{6}x)$$

$$3(7) - 2(5) = 22(2x)$$

$$21 - 10 = 44x$$

$$11 = 44x$$

$$\frac{11}{44} = x$$

$$\frac{1}{4} = x$$

2. Quadratic Equations

A **quadratic equation** is an equation containing a second-degree polynomial; for example

$$ax^2 + bx + c = 0$$

where a, b, and c are real numbers, and if $a \neq 0$, it is in standard form.

the square root property

With the x^2 term isolated, the square root property states that:

$$\text{if } x^2 = k, \text{ then } x = \pm\sqrt{k}$$

where k is a nonzero real number.

Example: Solve the quadratic using the square root property $x^2 = 8$

$$x^2 = 8$$
$$x = \pm\sqrt{8}$$
$$= \pm 2\sqrt{2}$$

The solutions are $2\sqrt{2}$ and $-2\sqrt{2}$.

Example: Factor and solve the equation $x^2 + x - 6 = 0$

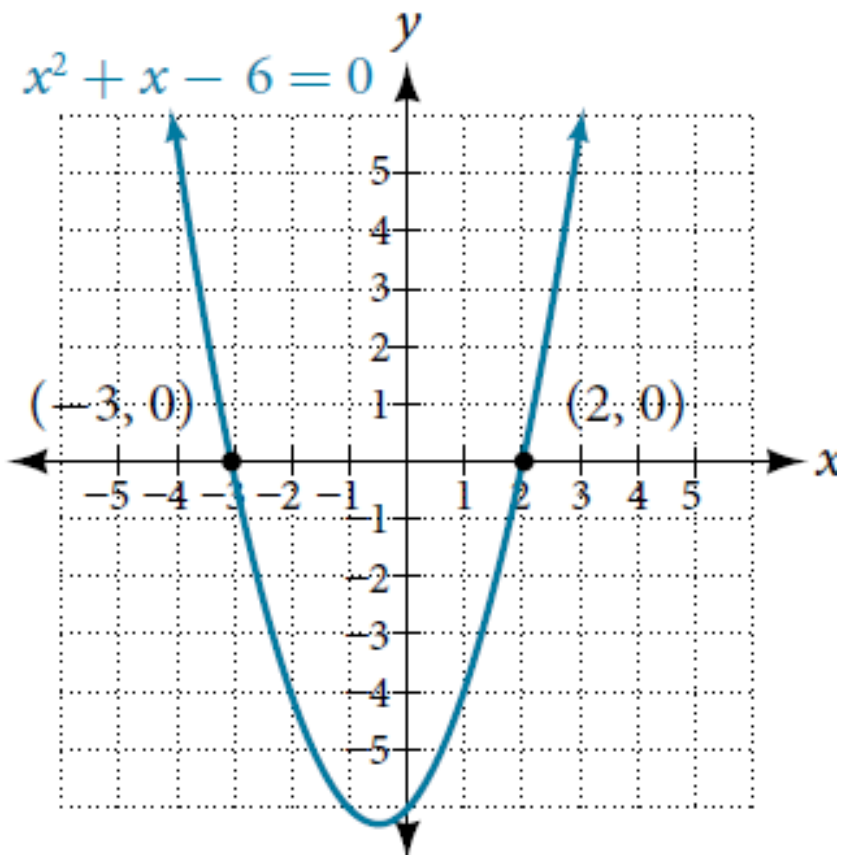
$$x^2 + x - 6 = (x - 2)(x + 3) = 0$$

$$(x - 2) = 0 \longrightarrow x = 2$$

$$(x + 3) = 0 \longrightarrow x = -3$$

The two solutions are 2 and -3

Product		Sum
1	-6	-5
-1	6	5
2	-3	-1
-2	3	1



Example: Solve the quadratic equation by factoring $x^2 + 8x + 15 = 0$

$$x^2 + 8x + 15 = (x + 3)(x + 5)$$

$$(x + 3) = 0 \longrightarrow x = -3$$

$$(x + 5) = 0 \longrightarrow x = -5$$

The solutions are -3 and -5.

Product		Sum
1	15	16
-1	-15	-16
3	5	8
-3	-5	-8

Solve the Quadratic Equation Using the Quadratic Formula

the quadratic formula

Written in standard form, $ax^2 + bx + c = 0$, any quadratic equation can be solved using the **quadratic formula**:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

where a, b, and c are real numbers and $a \neq 0$.

Example: Solve the quadratic equation: $x^2 + 5x + 1 = 0$

Identify the coefficients: $a = 1, b = 5, c = 1$. Then use the quadratic formula.

$$x = \frac{-(5) \pm \sqrt{(5)^2 - 4(1)(1)}}{2(1)}$$

$$= \frac{-5 \pm \sqrt{25 - 4}}{2}$$

$$= \frac{-5 \pm \sqrt{21}}{2}$$

Example: Use the quadratic formula to solve $x^2 + x + 2 = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(1) \pm \sqrt{(1)^2 - (4) \cdot (1) \cdot (2)}}{2 \cdot 1}$$

$$= \frac{-1 \pm \sqrt{1 - 8}}{2} = \frac{-1 \pm \sqrt{-7}}{2} = \frac{-1 \pm i\sqrt{7}}{2}$$

The solutions to the equation are $\frac{-1 + i\sqrt{7}}{2}$ and $\frac{-1 - i\sqrt{7}}{2}$

3. Radical Equations

radical equations

An equation containing terms with a variable in the radicand is called a **radical equation**.

Radical equations may have one or more radical terms, such as

$$\sqrt{3x+18} = x$$

$$\sqrt{x+3} = x-3$$

$$\sqrt{x+5} - \sqrt{x-3} = 2$$

Example: Solve $\sqrt{15-2X} = X$

$$\sqrt{15-2x} = x$$

$$\left(\sqrt{15-2x}\right)^2 = (x)^2$$

$$15-2x = x^2$$

Set it equal to zero and solve.

$$0 = x^2 + 2x - 15$$

$$0 = (x+5)(x-3)$$

$$0 = (x+5) \quad \text{or} \quad 0 = (x-3)$$

$$-5 = x \quad \text{or} \quad 3 = x$$

First, check -5.

$$\sqrt{15-2x} = x$$

$$\sqrt{15-2(-5)} = -5$$

$$\sqrt{25} = -5$$

$$5 \neq -5$$

Check 3.

$$\sqrt{15-2x} = x$$

$$\sqrt{15-2(3)} = 3$$

$$\sqrt{9} = 3$$

$$3 = 3$$

The solution is 3.

4. polynomial equations

polynomial equations

A polynomial of degree n is an expression of the type

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$$

where n is a positive integer and $a_n, \ldots, a_0$ are real numbers and $a_n \neq 0$.

Setting the polynomial equal to zero gives a **polynomial equation**. The total number of solutions (real and complex) to a polynomial equation is equal to the highest exponent n.

Example: Solve the polynomial by factoring: $5x^4 = 80x^2$

First, set the equation equal to zero. $5x^4 - 80x^2 = 0$

$$5x^2(x^2 - 16) = 0$$

$$5x^2 = 0 \qquad x^2 - 16 = 0$$

$$x = 0 \qquad (x - 4)(x + 4) = 0$$

$$x - 4 = 0 \quad \text{or} \quad x + 4 = 0$$

$$x = 4 \quad \text{or} \quad x = -4$$

The solutions are 0 (double solution), 4, and -4.